CHAMPS slides: exploratory data analysis

Generated 2026-11-08 from repository revision 99028f3 with `pixi run eda-report`. All numbers are descriptive counts of the shared dataset; no model was evaluated.

71,259	9,845	8	86%
whole-slide images	cases	sites	SVS slides linked to case data

Key observations

- 69% of slides are `.ndpi`; the SVS viewer and its stain and organ breakdowns cover only the 31% in `.svs`.
- Perinatal asphyxia/hypoxia is the underlying cause for 41% of cases with a DeCoDe result (per-site mix in section 2).
- Embedding clusters associate most with age group (Cramér's V 0.47).

1 · Slide inventory

The shared WSI inventory lists 71,259 slides from 9,845 cases (253 with a probe error). The SVS viewer covers the 22,263 `.svs` slides (4,262 cases); 19,168 of them link to case-level tables through the study-ID mapping.

Slides by site

Kenya	15,009	21%
Sierra Leone	13,046	18%
South Africa	10,498	15%
Mozambique	9,295	13%
Bangladesh	8,359	12%
Ethiopia	8,254	12%
Mali	5,585	8%
Nigeria Cross River	1,213	2%

File format

ndpi	48,996	69%
svs	22,263	31%

Objective power

40×	39,062	55%
20×	31,942	45%
unknown	254	0%
5×	1	0%

SVS slides by stain (filename)

HE	11,091	50%
No stain token	7,484	34%
IHC	2,748	12%
GRAM	504	2%
GMS	151	1%
TRICHROME	82	0%
IRON	64	0%
Other	139	1%

SVS slides by organ (slide registry)

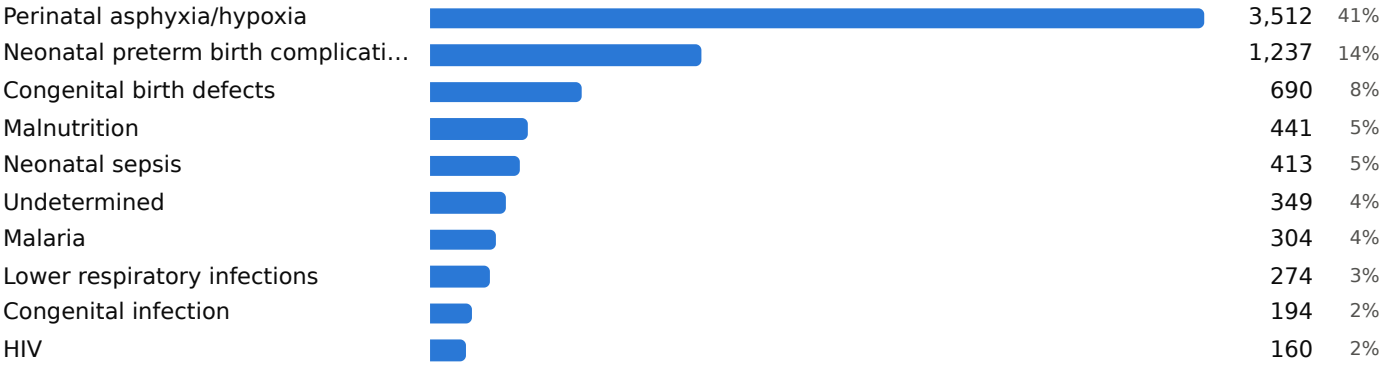
right lung	5,400	24%
left lung	4,674	21%
liver	3,968	18%
other	3,038	14%
cns post	2,593	12%
cns trans	2,249	10%
bone marrow	341	2%

Stain labels. Most site `.ndpi` slides carry no stain in the filename or metadata. A colour-feature classifier was explored (issue #5, single run) but could not separate special stains from H&E, so the pipeline now treats slides without a stain token as H&E (`bp26.inference.stains`). This is an assumption, not a measured label.

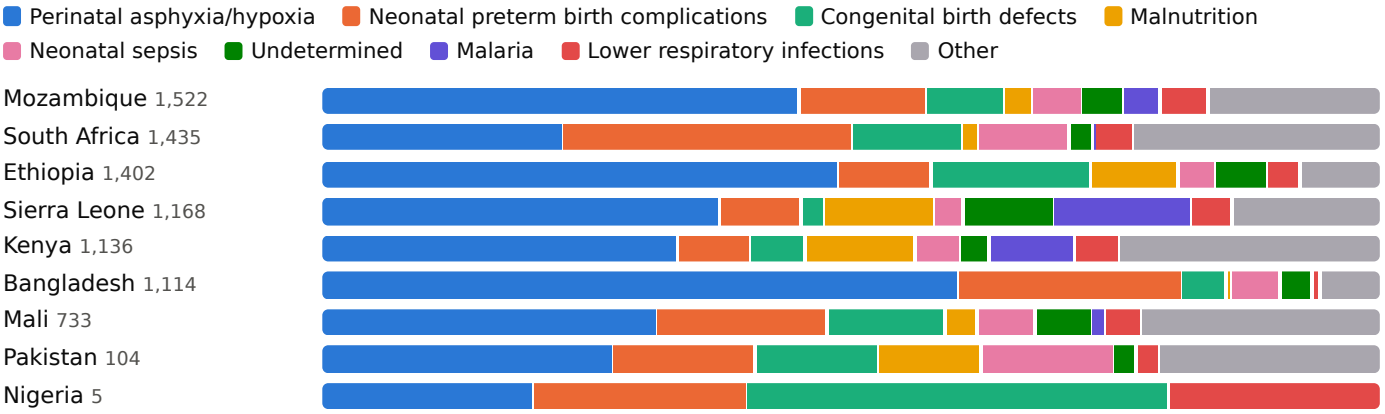
2 · Causes of death

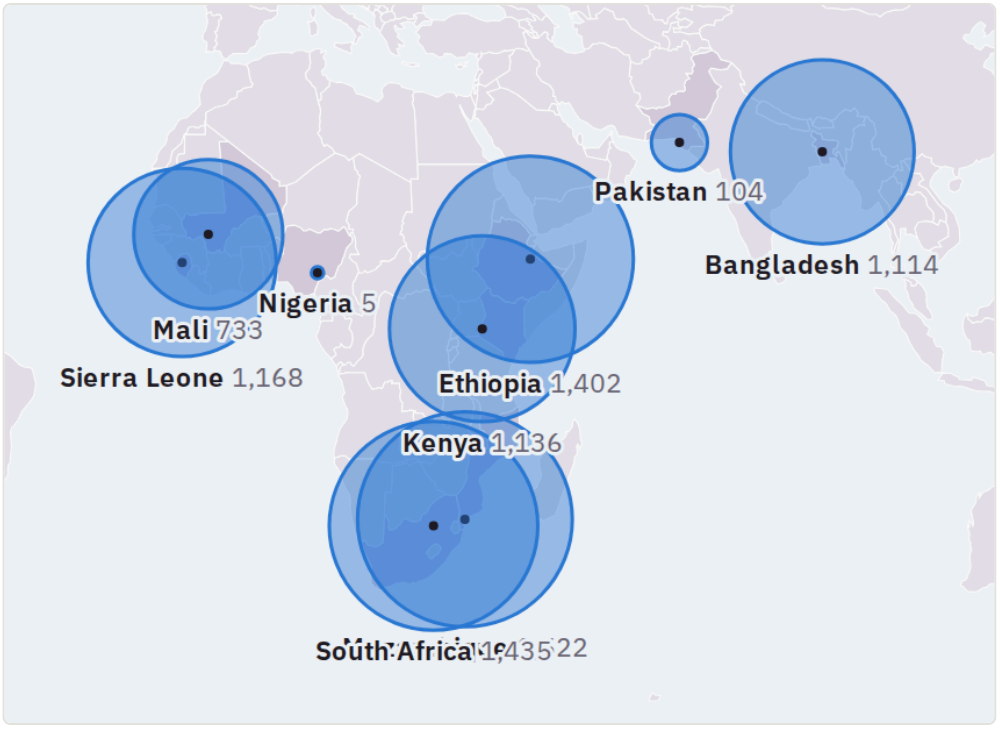
Underlying cause of death (DeCoDe) for 8,619 cases across 9 sites, grouped into 61 categories. Shares are of all cases with a DeCoDe result.

Top underlying causes



Mix of underlying causes per site (cases)

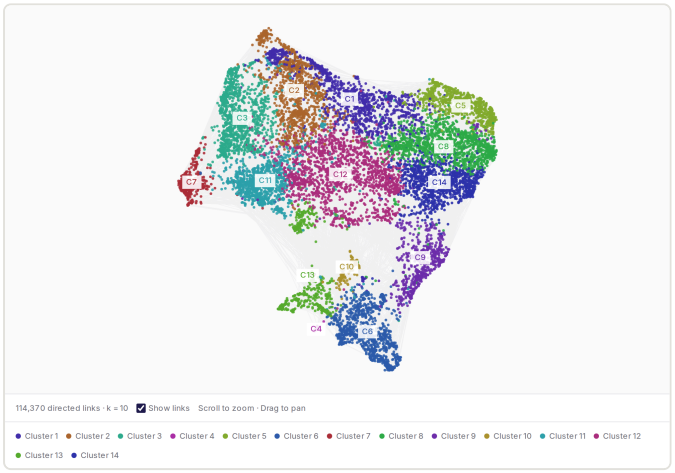




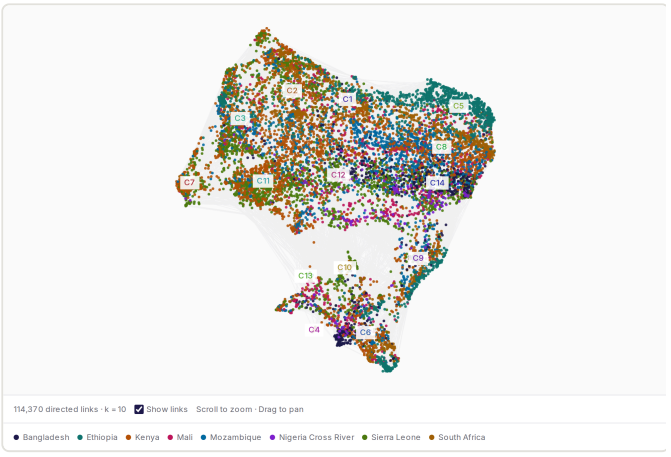
Disease map (SVS viewer): circle area is the site's share of all cases with a DeCoDe result.

3 · Slide embeddings

11,437 liver H&E slides embedded by the Virchow2 CLAM-MB model (finding branch “cholestasis”), projected with UMAP and clustered with Leiden into 14 clusters. Associations are computed per patient (7,671 patients; dominant cluster per patient).



Coloured by Leiden cluster.



Coloured by site.

Cluster–property associations (95% patient bootstrap intervals)

Property	Metric	Effect	AMI	Patients with value
Site	Cramér's V	0.314 [0.308, 0.324]	0.146	7,671
Age group	Cramér's V	0.466 [0.459, 0.475]	0.268	7,383
Sex	Cramér's V	0.049 [0.048, 0.079]	0.001	7,381
Age (months)	η^2	0.405 [0.377, 0.432]	—	7,383

Embeddings come from a supervised model and include its training slides, so clusters reflect what that model learned, not held-out structure. Associations are descriptive, not causal.